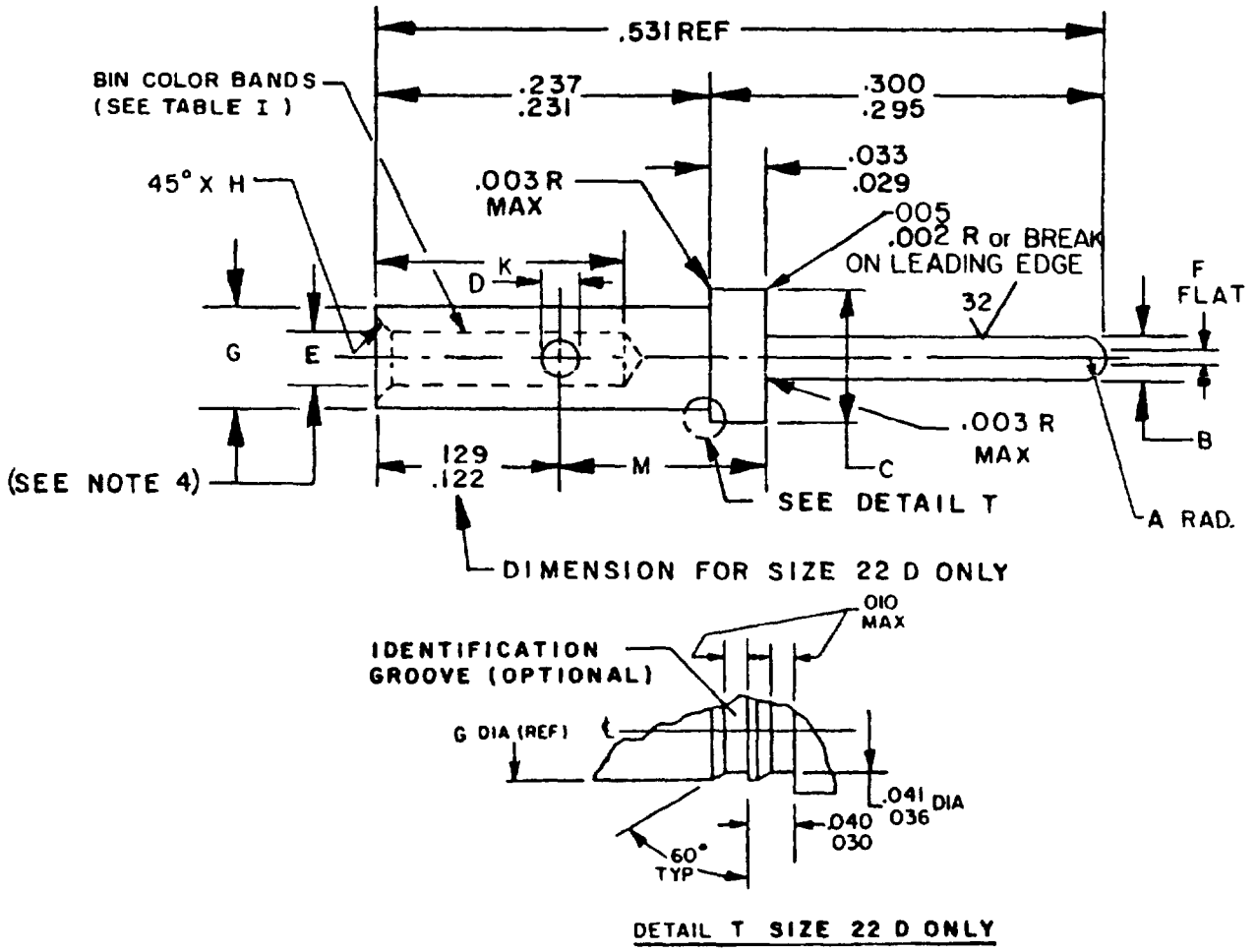
MILITARY SPECIFICATION SHEET

CONTACTS, ELECTRICAL CONNECTOR, PIN, CRIMP REMOVABLE, THERMOCOUPLE, (FOR MIL-C-38999 SERIES I, II, III, AND IV, MIL-C-24308, AND MIL-C-83733)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the contacts described herein shall consist of this specification and the latest issue of MIL-C-39029.



DETAIL T SIZE 22 D ONLY

FIGURE 1. Pin contact.

© denotes change

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil - Downloaded: 2026-10-21T18:48:32
Check the source to verify that this is the current version before use.

Inches	mm	Inches	mm	Inches	mm
.002	0 05	.032	0.81	.070	1.78
.003	0.08	.033	0.84	072	1.83
005	0.13	.0335	0 85	.078	1 98
.010	0.25	.0355	0 90	082	2 08
011	0.28	036	0 91	.088	2.24
.015	0 38	039	0.99	091	2 31
.018	0.46	.041	1.04	.094	2 39
.020	0.51	.042	1 07	101	2 57
022	0 56	046	1 17	.103	2.62
025	0.64	.048	1.22	127	3 23
026	0 66	060	1 52	130	3.30
.029	0.74	0615	1 56	141	3 58
.0295	0 75	062	1.57	.157	3 99
.030	0.76	.0635	1 61	209	5 31
0305	0 77	066	1.68	229	5 82
		068	1.73	531	13 46

BIN Code	A Radius	B Dia	C Dia	D Dia	E Dia	F Dia	G Dia	H	K	M
470 thru 473	.020 .010	.0305 .0295	.062 .060	.022 .018	.0355 .0335	.011 Max	.048 .046	.005 .003	.157 .141	---- ----
474 thru 477	.020 .015	.041 .039	.094 .091	.032 .026	.048 .046	.015 Max	.070 .068	.010	.229 .209	.078 .072
478 thru 481	.025 .020	.0635 .0615	.130 .127	.042 .036	.068 .066	.030 .011	.103 .101	.005	.229 .209	.088 .082

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.
4. For size 22D only, diameters E and G are to be concentric within .003 (TIR) regardless of feature size (RFS), for all other contact sizes, diameters E and G to be concentric within .001 (TIR) of maximum material condition (MMC).
5. Spherical end dimensions shall conform to MIL-C-39029/58.

FIGURE 1. Pin contacts - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:43Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Qualification: Contacts shall comply with reliability assurance provisions of MIL-STD-790 as specified in MIL-C-38999.

Dimensions, design characteristics, and configuration See figure 1 and table I.

Mating contacts: MIL-C-39029/88 and MIL-C-39029/89.

Tools: See table II.

Materials and plating: See table III.

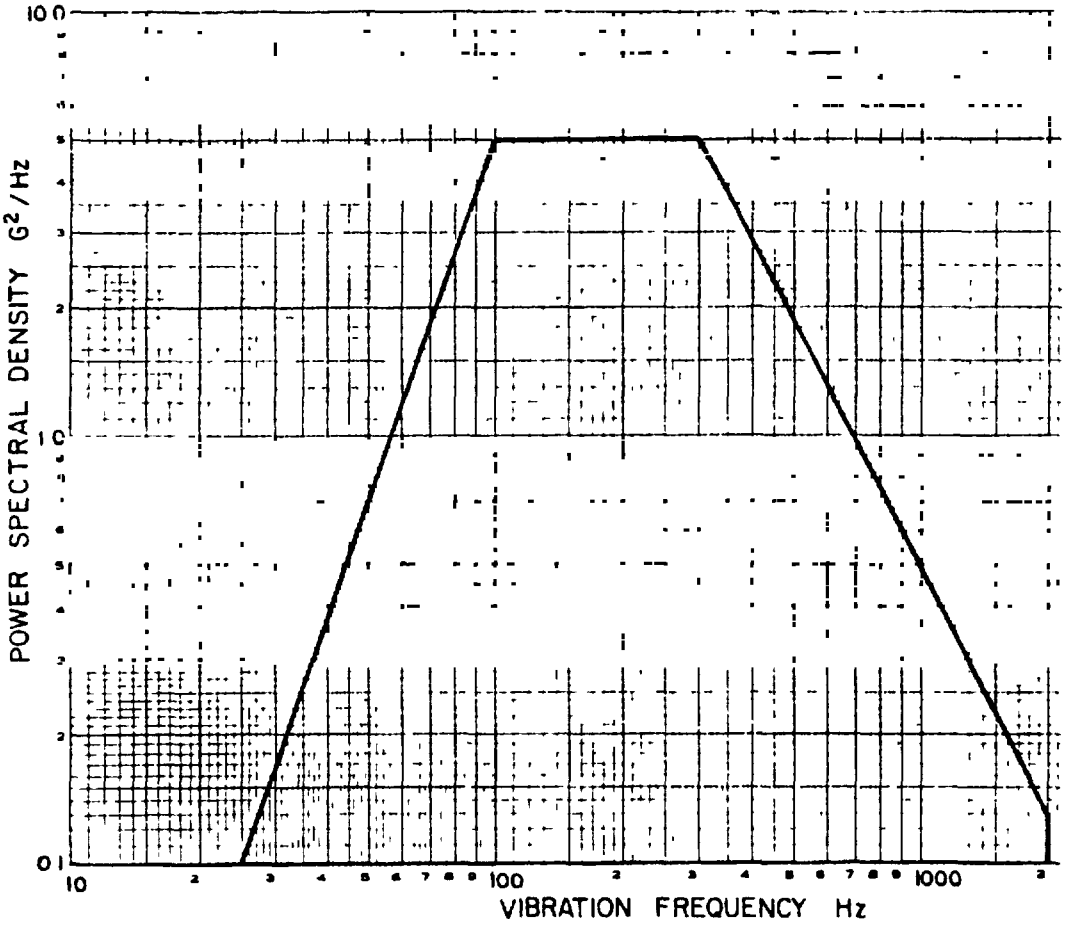
Tensile strength See table III.

Random vibration. Connectors shall be subjected to the test specified in method 2005 of MIL-STD-1344. The following details shall apply:

a. Test condition V using the vibration envelope shown below. (Derived from zone 2 outlined in Aerospace Information Report AIR 1557.)

b. Vibration to be conducted at standard test conditions

c. Duration shall be 8 hours in the longitudinal direction and 8 hours in a perpendicular direction for a total of 16 hours.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:43Z
Check the source to verify that this is the current version before use.

TABLE I. Design characteristics.

BIN code	Color bands			Mating end size	Wire barrel size	Type	Class
	1st.	2nd.	3rd.				
470	Yellow	Violet	Black	22	22D	C (See table III for composition)	B
471	Yellow	Violet	Brown	22	22D		
472	Yellow	Violet	Red	22	22D		
473	Yellow	Violet	Orange	22	22D		
474	Yellow	Violet	Yellow	20	20		
475	Yellow	Violet	Green	20	20		
476	Yellow	Violet	Blue	20	20		
477	Yellow	Violet	Violet	20	20		
478	Yellow	Violet	Gray	16	16		
479	Yellow	Violet	White	16	16		
480	Yellow	Gray	Black	16	16		
481	Yellow	Gray	Brown	16	16		

TABLE II. Tools.

BIN code	Basic crimping tool	Positioner	Installing tool	Removal tool
470	M22520/2-01	M22520/2-09	M81969/14-01	M81969/14-01
471	M22520/7-01	M22520/7-07	M81969/8-01	M81969/8-02
472				
473				
474	M22520/2-01	M22520/2-10	M81969/8-05	M81969/8-06
475	M22520/1-01	M22520/1-04	M81969/14-02	M81969/14-02
476	M22529/7-01	M22520/7-08		
477				
478	M22520/1-01	M22520/1-04	M81969/8-07	M81969/8-08
479	M22520/7-01	M22520/7-04	M81969/14-03	M81969/14-03
480				
481				

Ⓒ TABLE III. Materials, plating, and tensile strength.

BIN code	Materials	Plating	Tensile strength 1/			
			Wire size	Axial load (lbf)	Wire size	Axial load (lbf)
470	Constantan	None	28 min	1.5	22 max	6
471	Alumel	None				
472	Chromel	None				
473	Iron	Cadmium plate 2/				
474	Constantan	None	24 min	4.5	20 max	14
475	Alumel	None				
476	Chromel	None				
477	Iron	Cadmium plate 2/				
478	Constantan	None	20 min	14	16 max	33
479	Alumel	None				
480	Chromel	None				
481	Iron	Cadmium plate 2/				

1/ Applies to type 1 wire per MIL-W-5846.
2/ Chromate clear coat.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:43Z
Check the source to verify that this is the current version before use.

High-impact shock: Connectors shall be coupled together by normal coupling means. All connectors shall be wired in a series circuit with 100 milliamperes maximum current flow through the series circuit during high-impact shock. Connectors shall be monitored for any discontinuities. A detector capable of detecting all discontinuities in excess of 1 microsecond shall be used. Wired and mated connectors shall be subjected to the test specified in MIL-S-901, grade A with the following modifications and additions. Mounting fixture shall be in accordance with MIL-S-901, light weight. The cable or wire bundle shall be supported on a stationary frame in such a manner to provide a free flexing cable length between frame and fixture of not less than 36 inches (914.4 mm).

Test condition A. The plug shall be terminated with at least 80 percent of wired contacts. The wire bundle shall be provided with straight, open frame, strain relief accessory hardware.

QPL evaluating activity: Defense Electronics Supply Center (DESC-E), Dayton, Ohio, 45444.

International interest: NEPR 57.

Part numbers: See table IV.

TABLE IV. Part number and BIN code.

BIN code	Part number	Superseded
470	M39029/87-470	M39029/872222C1
471	M39029/87-471	M39029/872222C2
472	M39029/87-472	M39029/872222C3
473	M39029/87-473	M39029/872222C4
474 1/	M39029/87-474	M39029/872020C1
475 1/	M39029/87-475	M39029/872020C2
476 1/	M39029/87-476	M39029/872020C3
477 1/	M39029/87-477	M39029/872020C4
478 1/	M39029/87-478	M39029/871616C1
479 1/	M39029/87-479	M39029/871616C2
480 1/	M39029/87-480	M39029/871616C3
481 1/	M39029/87-481	M39029/871616C4

1/ Not for use with MIL-C-83733 or MIL-C-24308 connectors (for MIL-C-83733 contact sizes 20 and 16, see MIL-C-39029/9 and MIL-C-39029/10).

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities.
Army - AR
Navy - EC
Air Force - 11, 99
DLA - ES

User activities:
Navy - OS, SH

Preparing activity
Air Force - 85

Agent:
DLA - ES

(Project 5935-3586-3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:43Z
Check the source to verify that this is the current version before use.